Cognizant

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Cognizant Academy

Prince Mobiles

C#, ADO .Net Core Knock Out

ChallengeVersion 1.0

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Cognizant

Table Of Contents

1.0 Introduction **3**

1.1 Purpose of this document **3**

1.2 Definitions & Acronyms **3**

1.3 Project Overview **3**

1.4 Scope **3**

1.5 Target Audience **4**

1.6 Software Requirements **4**

2.0 Functional Requirements **4**

2.1 Functional Requirements **4**

2.2 Use case Diagram **6**

2.3 System Architecture Diagram **6**

2.4 Sample Input/Output **7**

3.0 Design Specification **9**

3.1 Data Design **9**

3.2 Component Details for identified Use Cases **10**

3.2.1 Get data and update in the database 10

3.2.2 Remove Mobile Details By Model Name 10

3.2.3 Calculate the Net Amount 11

3.3 Component Specification **11**

3.3.1 UpdateMobilePrice Method 12

3.3.2 RemoveMobileDetailsByModelName Method 12

3.3.3 NetAmountCalculation Method 13

3.3.4 GetConnection Method 14

3.4 General Design Constraints **14**

4.0 Submission **14**

4.1 Code submission instructions 14

5.0 Change Log **15**

6.0 Evaluation Areas **15**

1.0 Introduction

1.1 Purpose of this document

Prince Mobiles is a mobile sales and service company. They have a wide range of products and services to meet the varied requirements of their customers. They provide some discount offers for their customers, so they need an application to update and remove the mobile details and to calculate the net amount after the discount.

Prince Mobile's current focus is to automate the below mentioned requirements.

1. Update the price details by IME number in the database.

2. Delete the mobile details by model name from the database.

3. Calculate the net amount for the particular Manufacturer.

1.2 Definitions & Acronyms

Definition / Acronym	Description
Req	Requirement

1.3 Project Overview

This project captures the various concepts, techniques and skills learned and helps to put them into practice using C# with ADO.NET. Admittedly, this would be at a scaled-down level since the purpose is to let the associate experience the various concepts learned in C# as an individual. The individual associate is expected to create a console-based application within the specified time.

1.4 Scope

The scope of the system is explained through its following modules.

1. Update the price details by IME number in the database.
2. Delete the mobile details by model name from the database.
3. Calculate the net amount for the particular Manufacturer.

1.5 Target Audience

Learner Level

1.6 Software Requirements

#	Item	Specification/Version
1.	C#	10
2.	ADO.NET	6.0
3	SQLSERVER	2017

Note: All the required software will be provided in the Tekstac platform.

2.0 Functional Requirements

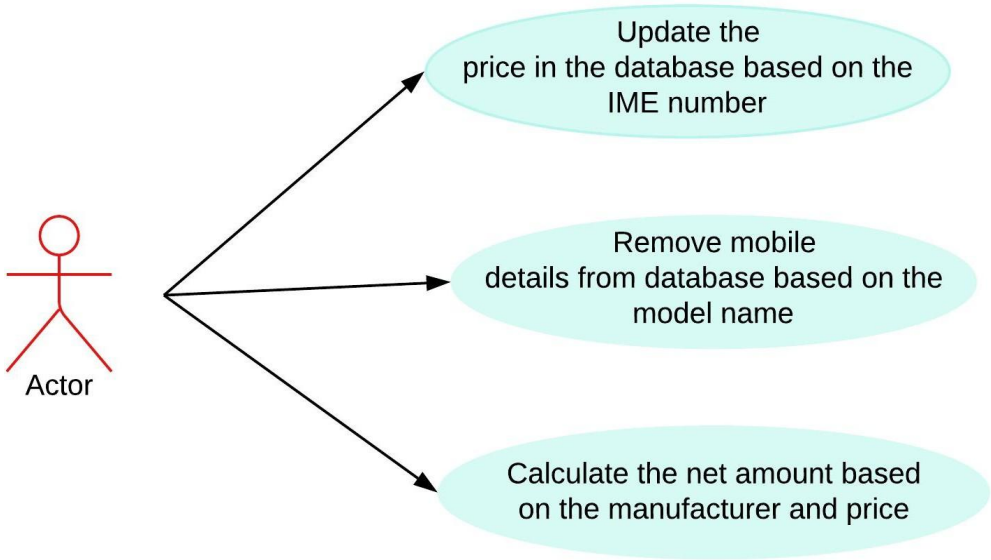
2.1 Functional Requirements

Req. #	1
Req. Name	Update the mobile price details in the database based on the given IME number.
Req. Description	The IME number and price to be updated are entered through the Main method. Update the mobile details in the database.
Actors/ Users	Admin
Comments	The admin is responsible for updating the data .

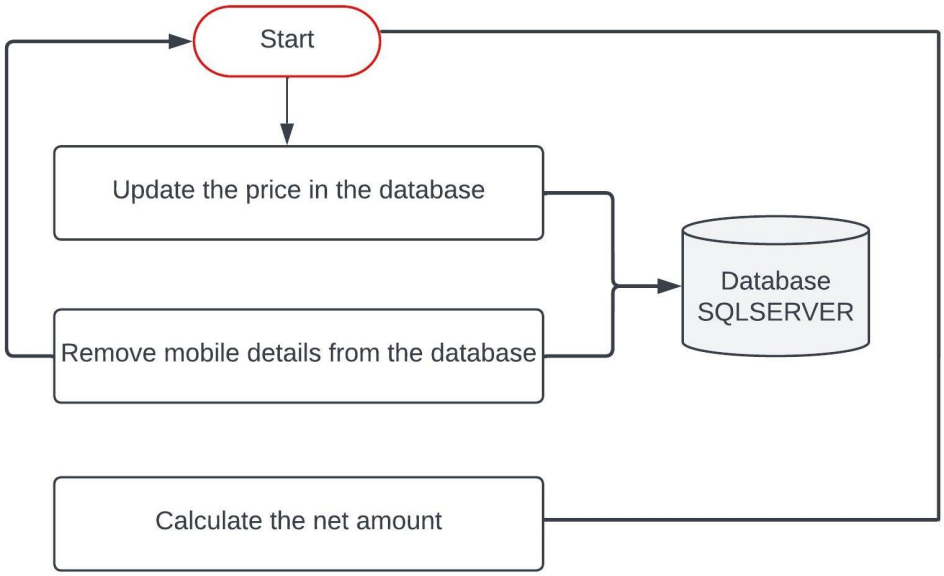
Req. #	2
Req. Name	Remove all the mobile details based on the mobile model name.
Req. Description	The user enter the mobile model name and it will delete all the details from the database
Actors/ Users	Admin
Comments	The admin is responsible for deleting the data

Req. #	3
Req. Name	Calculate the Net Amount
Req. Description	Calculate the net amount with the help of the Manufacturer and price.
Actors/ Users	Users
Comments	User is Responsible for calculating the Net Amount.

2.2 Use case Diagram



2.3 System Architecture Diagram



2.4 Sample Input/Output

1. Update the mobile price

2. Remove mobile details by model name

3. Calculate the net amount

4. Exit

Enter your choice

1

Enter the IME number

NO101

Enter the price to be update

9600

Price updated successfully

1. Update the mobile price

2. Remove mobile details by model name

3. Calculate the net amount

4. Exit

Enter your choice

2

Enter the mobile model name

Nokia C30

Mobile details removed successfully

1. Update the mobile price
2. Remove mobile details by model name
3. Calculate the net amount
4. Exit

Enter your choice

3

Enter the manufacturer name

Samsung

Enter the price

15000

The mobile net amount is : 14250

1. Update the mobile price
2. Remove mobile details by model name
3. Calculate the net amount
4. Exit

Enter your choice

1

Enter the IME number

12345

Enter the price to be update

15500

Price updates failed

```
1.Update the mobile price

2.Remove mobile details by manufacturer

3.Calculate the net amount

4.Exit

Enter your choice

4

Thank you
```

3.0 Design Specification

3.1 Data Design

Table Structure:

Table name: Mobile_Master	
Column Name	**Data type**
IME_No	varchar(10)– Primary Key
Model_Name	varchar(20)
Manufacturer	varchar(20)
Warranty_In_Years	numeric(10)
Price	numeric(7,2)

Design Constraints:

- Use **SQLSERVER** database to store the data. The database name is "SalesDatabase".**This is already created for you in Tekstac.**
- The table names and the column names should be the same as specified in the table structure.
- The database connection information is specified in the "appsettings.json" file, which is also provided as part of code skeleton. **THIS IS GIVEN ONLY FOR YOUR REFERENCE**. You need **NOT** change this. The code skeleton will be available in the Tekstac platform.

Sample Data is Already inserted in the Mobile_Master table.

IME_No	Model_Name	Manufacturer	Warranty_In_Years	Price
NO101	Nokia C30	Nokia	2	15999.00
NO102	Nokia XR20	Nokia	3	41998.00
SA201	Galaxy A72	Samsung	3	36999.00
SA202	Galaxy M13	Samsung	2	9999.00
SO301	Xperia XZ1	Sony	3	39433.00
SO302	Xperia X	Sony	2	18499.00

Note: The code skeleton will be available in the Tekstac platform

3.2 Component Details for identified Use Cases

3.2.1 Get data and update in the database

In the 'Main' get the following value

IME_No Price

3.2.2 Delete Mobile Details By Manufacturer

In the 'Main' get the following value,

Manufacturer

3.2.3 Calculate the Net Amount

Calculate the Net Amount Based on the Manufacturer and Price. Get the Manufacturer and Price from the user in 'Main' as shown in the sample input/output.

For sample input/output, refer section 2.4.

Calculate the Net Amount based on the below

condition:

NOTE : Manufacturer can be **Nokia**, **Samsung** or

Sony (**Case Sensitive**)

Calculating Net Amount

Manufacturer	**Price Range**	**Discount**
Nokia	Upto 20000	10% of Price
	Above 20000	20% of Price
Samsung	Upto 20000	5% of Price
	Above 20000	15% of Price
Sony	Upto 20000	3% of Price
	Above 20000	13% of Price

3.3 Component Specification

Class Name : Mobile

Create a Class **Mobile** with the following public Properties.

Data Type	Variable Name
string	IMENumber
string	ModelName
string	Manufacturer
int	Warranty
double	Price

Note: Keep all the properties **'public'.**

Class Name : MobileUtility

Create a Class **MobileUtility** to use the below methods and functions:

3.3.1 UpdateMobilePrice Method

Type(Class)	Method	Responsibilities
MobileUtility	public bool UpdateMobilePrice(string imeNumber, double price)	This method is used to update the mobile price based on the corresponding IME number. If the IME number is available in the database table then update the price and return **true,** Otherwise return **false.** If this method returns true, then display **Price updated successfully** in the Main method.

		If this method returns false, then display **Price updates failed** in the Main method.

3.3.2 RemoveMobileDetailsByModelName Method

Type(Class)	Method	Responsibilities
MobileUtility	public bool RemoveMobileDetails(string modelName)	This method is used to delete the mobile details based on the mobile model name. If the mobile details are deleted, then return **true** and display **Mobile details removed successfully** in the Main method

3.3.3 NetAmountCalculation Method

Type(Class)	Method	Responsibilities
MobileUtility	public double NetAmountCalculation(string manufacturer,double price)	This method is used to calculate the net amount based on the given manufacturer and price. If the manufacturer is invalid then return **the same price**.

Class Name : DBConnection

3.3.4 GetConnection Method

Responsibility:

The below given method is already provided. Your task is to return the SqlConnection String from this method.

Type(Class)	Method	Resources
DBConnection	public SqlConnection GetConnection()	This method is used to connect to SQLSERVER database. Use the connection name "DBConnection" in appsettings.json to create a connection.

Note : To create the database, navigate to the "**Follow the instructions**" folder within the Solution Explorer. Open the **instruction.txt** file and carefully follow the guidelines provided. Execute the queries specified in the **script.txt** file to populate the database with the necessary information.

3.4 General Design Constraints

1. The properties/method/class name should be correctly specified as given in the document.
2. Keep all the classes as '**public**'
3. Do not change the **appsettings.json** file.
4. Do not change the namespace name.

4.0 Submission

4.1 Code submission instructions

1. Do not change the code skeleton given, as your code will be auto evaluated.
2. Your last submitted solution will be considered for

detailed evaluation.

3. Make sure to submit the solution before the time limit. After the assessment duration you will not be allowed to submit the solution.

5.0 Change Log

	Changes Made
V1.0.0	Initial baseline created on <dd-Mon-yy> by <Name of Author>
Vx.y.z	<Please refer the configuration control tool / change item status form if the details of changes are maintained separately. If not, the template given below needs to be followed>

Section No.	Changed By	Effective Date	Changes Effected

6.0 Evaluation Areas

S.No	Description
1.	Declaration of properties and methods in the class Mobile and MobileUtility.
2.	Declaration of methods in the class DBConnection
3.	Implementation to create a valid database connection object
4.	Implementation to update the price details in the database
5.	Implementation to delete mobile details by mobile model name from the database.

6.	Calculation of Net Amount by manufacturer and price.